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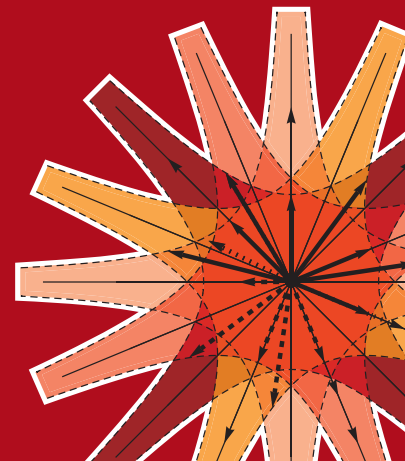
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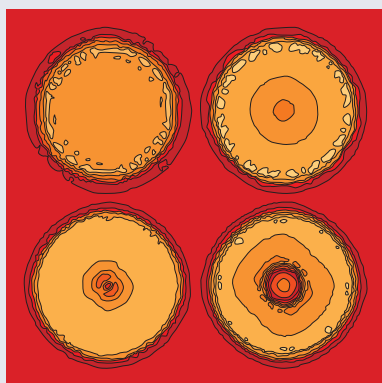
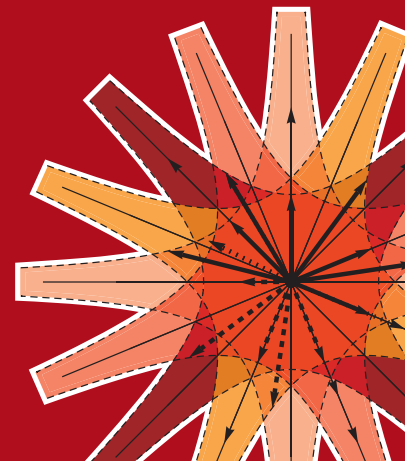


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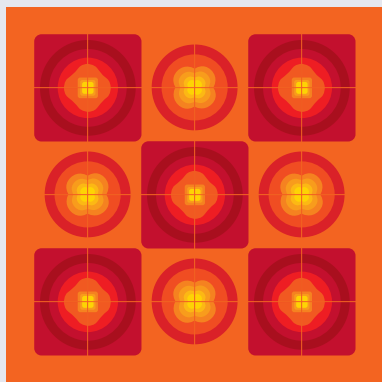
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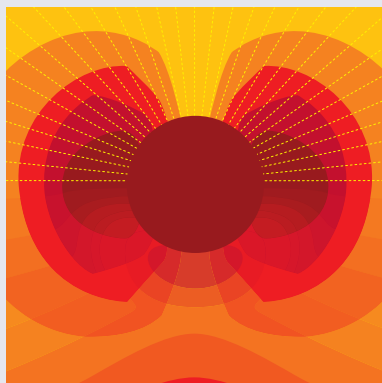
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Cooling a nanomechanical resonator by a triple quantum dot

ZENG-ZHAO LI^{1,2}, SHI-HUA OUYANG^{1,2}, CHI-HANG LAM² and J. Q. YOU^{1(a)}

¹ Department of Physics and State Key Laboratory of Surface Physics, Fudan University - Shanghai 200433, China

² Department of Applied Physics, Hong Kong Polytechnic University - Hung Hom, Hong Kong, China

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Abstract – We propose an approach for achieving ground-state cooling of a nanomechanical resonator (NAMR) capacitively coupled to a triple quantum dot (TQD). This TQD is an electronic analog of a three-level atom in Λ configuration which allows an electron to enter it via lower-energy states and to exit only from a higher-energy state. By tuning the degeneracy of the two lower-energy states in the TQD, an electron can be trapped in a dark state caused by destructive quantum interference between the two tunneling pathways to the higher-energy state. Therefore, ground-state cooling of an NAMR can be achieved when electrons absorb readily and repeatedly energy quanta from the NAMR for excitations.

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Introduction. – Nanomechanical resonators (NAMRs) with high resonance frequencies and large quality factors are currently attracting considerable attention owing to their wide range of potential applications (see, *e.g.*, [1,2]). Moreover, quantized NAMRs are potentially useful for quantum information processing. For example, quantized motion of buckling nanoscale bars has been proposed for qubit implementation [3] and also for creation of quantum entanglement [4–6]. However, the dynamics of the NAMRs must approach the quantum regime, which is usually difficult to arrive at due to interactions with other components as well as the environments.

One way to achieve the quantum regime for an NAMR is to increase the resonance frequency so that an energy quantum of the NAMR is larger than the thermal energy. Recently, NAMRs based on metallic beams [7] and carbon nanotubes [8] with resonance frequencies about several hundred megahertz have been developed. However, a temperature lower than 10 mK (below the typical dilution refrigerator temperature) is required for an NAMR with a frequency of 200 MHz to operate in the quantum regime. Therefore, one still needs to cool the NAMR further, for instance via coupling to an optical or an on-chip electronic system. Various experiments on the cooling of a single NAMR via radiation pressure or dynamical

backaction have been reported (see, *e.g.*, [9–15]). Other cooling mechanisms based on a Cooper pair box [16] or a three-level flux qubit [17] with periodic resonant coupling have also been theoretically proposed. In these approaches, a strong resonant coupling between the NAMR and the qubit is required for the NAMR to achieve its ground-state cooling.

In the weak coupling regime, a conventional method for cooling an NAMR is the sideband cooling approach (see, *e.g.*, refs. [18–23]), where an NAMR is usually coupled to a two-level system (TLS) with the two states being electronic states in quantum dots [18–20], photonic states in a cavity [21–23], or charge states in superconducting circuits [24]. In order to achieve ground-state cooling, the resolved-sideband cooling condition $\omega_m \gg \Gamma$ (with ω_m denoting the oscillating frequency of the NAMR and Γ the decay rate of the TLS) must be followed to selectively drive the lowest sideband of the TLS. However, the frequency of a typical NAMR is about 100 MHz [7,8] which is in general of the same order of the decay rate of the two-level system, indicating that the resolved-sideband cooling condition is not easy to be fully fulfilled.

In this letter, we propose a different approach to cool an NAMR, in the non-resolved-sideband cooling regime, via quantum interference in a capacitively coupled triple quantum dot (TQD) (see fig. 1(a)). Here we focus on the strong Coulomb-blockade regime with at most one electron being allowed in the TQD at one time. The TQD acts

^(a)E-mail: jqyou@fudan.edu.cn

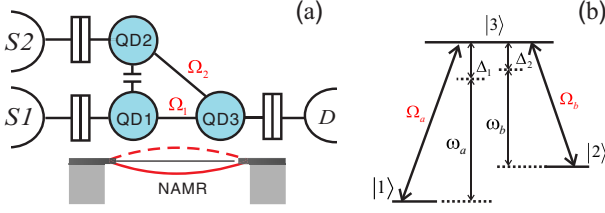


Fig. 1: (Color online) (a) Schematic diagram of a TQD system. Dots 1 and 2 are both tunnel-coupled to dot 3 (with interdot coupling strengths Ω_1 and Ω_2 , respectively) while they are only capacitively coupled to each other. An NAMR is capacitively coupled to dots 1 and 3 of the TQD. (b) A three-level system in Λ configuration. The energy detuning of the laser frequency ω_a (ω_b) from the transition frequency between state $|1\rangle$ ($|2\rangle$) and state $|3\rangle$ is Δ_1 (Δ_2). The corresponding driving strength of the laser field is Ω_a (Ω_b).

as a three-level system in Λ configuration (see fig. 1(b)), in which the two dot states $|1\rangle$ and $|2\rangle$ (*i.e.*, the single-electron orbital states in dots 1 and 2) are coupled to a third (excited) state $|3\rangle$ via two tunnel barriers. We will show that a dark state in the TQD can be obtained by properly tuning the gate voltages to make the two lower-energy states degenerate. This dark state was previously proposed in refs. [25] and [26] to explain the vanishing of the charge current transporting through a TQD system. Here, capacitively coupled to the TQD, the NAMR can be cooled to its ground state, in analogy to the slowing down of the atoms via quantum interference [27]. Our approach has the following potential advantages: i) The cooling system presented here is completely electronic and thus can be conveniently fabricated on a chip. ii) By adjusting simply the gate voltages, the two degenerate lower-energy states, required for destructive quantum interference, are easily achieved. iii) In contrast to the cooling of an NAMR by coupling it to a superconducting qubit [28] or time-dependent optical cavities [29], the decay rate Γ of the higher-energy state of the TQD, which is equal to the rate of electron tunneling from the TQD to the right electrode, is tunable by varying the gate voltage.

Model. – The device layout of an NAMR coupled to a TQD is shown in fig. 1(a). The TQD is connected to three electrodes via tunnel barriers. In the TQD, dots 1 and 2 are only capacitively coupled to each other without electrons tunneling directly between them. In contrast, electrons can tunnel between dots 1 and 3 as well as between dots 2 and 3. Such capacitively coupled dots have already been achieved in experiments (see, *e.g.*, [30]). In the strong Coulomb-blockade regime, four electronic states need to be considered, *i.e.*, the vacuum state $|0\rangle$, and states $|1\rangle$, $|2\rangle$ and $|3\rangle$ corresponding to a single electron in the respective dot. The NAMR is capacitively coupled to dots 1 and 3 as shown in fig. 1(a).

The total Hamiltonian of the whole system is

$$H_{\text{total}} = H_0 + H_{\text{int}} + H_{\text{T}} + H_{\text{ep}}. \quad (1)$$

The unperturbed Hamiltonian H_0 is defined as $H_0 = H_{\text{TQD}} + H_{\text{leads}} + H_{\text{R}} + H_{\text{ph}}$, where

$$H_{\text{TQD}} = -\Delta_1 a_1^\dagger a_1 - \Delta_2 a_2^\dagger a_2 + (\Omega_1 a_1^\dagger a_3 + \Omega_2 a_2^\dagger a_3 + \text{H.c.}), \quad (2)$$

$$H_{\text{leads}} = \sum_{i,k} E_{ik} c_{ik}^\dagger c_{ik}, \quad (3)$$

$$H_{\text{R}} = \omega_m b^\dagger b, \quad (4)$$

$$H_{\text{ph}} = \sum_q \omega_q b_q^\dagger b_q \quad (5)$$

are Hamiltonians of the TQD, the electrodes, the NAMR and the thermal bath, respectively. We have put $\hbar = 1$. The energy of state $|3\rangle$ is chosen as the zero-energy point and $-\Delta_{1(2)}$ is the energy of state $|1\rangle$ ($|2\rangle$) relative to state $|3\rangle$. $a_i^\dagger$ creates an electron in the i -th dot ($i = 1, 2$ or 3) and $c_{ik}^\dagger$ (c_{ik}) is the creation (annihilation) operator of an electron with momentum k in the i -th electrode. The phonon operators $b^\dagger$ and b create and annihilate an excitation of frequency ω_m in the NAMR, respectively. The thermal bath is modeled as a bosonic bath with $b_q^\dagger$ (b_q) being the bosonic creation (annihilation) operator at frequency ω_q .

The electromechanical coupling between the NAMR and dots 1 and 3 of the TQD is given by

$$H_{\text{int}} = -(g_3 a_3^\dagger a_3 + g_1 a_1^\dagger a_1)(b^\dagger + b), \quad (6)$$

with a coupling strength $g_i = \eta_i \omega_m$ ($i = 1, 3$). For simplicity we consider $g_3 = g_1 = g$ (or $\eta_3 = \eta_1 = \eta$). For a typical electromechanical coupling, $\eta \sim 0.1$ (see, *e.g.*, ref. [31]). The tunneling coupling between the TQD and the electrodes is

$$H_{\text{T}} = \sum_{ik} (\Omega_{ik} a_i^\dagger c_{ik} + \text{H.c.}), \quad (7)$$

where Ω_{ik} characterizes the coupling strength between the i -th dot and the associated electrode via a tunnel barrier. Moreover, the coupling of the NAMR to the outside thermal bath is characterized by

$$H_{\text{ep}} = \sum_q \Omega_q (b_q^\dagger b + \text{H.c.}), \quad (8)$$

with Ω_q being the coupling strength.

Quantum dynamics of the triple quantum dot. –

Analogy between a TQD and a Λ -type three-level atom.

We now show that in the absence of the NAMR, the TQD system is analogous to a typical Λ -type three-level atom in the presence of two classical electromagnetic fields (see fig. 1(b)). The Hamiltonian of the field-driven Λ -type three-level system can be written as $H_\Lambda = \omega_1 a_1^\dagger a_1 + \omega_2 a_2^\dagger a_2 + \omega_3 a_3^\dagger a_3 + \Omega_a \cos(\omega_a t)(a_1^\dagger a_3 + a_3^\dagger a_1) + \Omega_b \cos(\omega_b t) \times (a_2^\dagger a_3 + a_3^\dagger a_2)$, where ω_i ($i = 1, 2$ or 3) is the energy of the i -th state in the three-level system; ω_a and ω_b are the

frequencies of the two driving fields with Ω_a and Ω_b being the corresponding driving strengths. In order to eliminate the time dependence of the Hamiltonian, we transform the system into a rotating frame defined by $U_R = e^{iRt}$ with $R = \omega_a a_1^\dagger a_1 + \omega_b a_2^\dagger a_2 - \omega_3(a_1^\dagger a_1 + a_2^\dagger a_2 + a_3^\dagger a_3)$. Within the rotating-wave approximation, the Hamiltonian becomes

$$\tilde{H}_\Lambda = -\Delta_1 a_1^\dagger a_1 - \Delta_2 a_2^\dagger a_2 + \Omega_1(a_1^\dagger a_3 + a_3^\dagger a_1) + \Omega_2(a_2^\dagger a_3 + a_3^\dagger a_2), \quad (9)$$

where $\Delta_1 = \omega_3 - \omega_1 - \omega_a$ and $\Delta_2 = \omega_3 - \omega_2 - \omega_b$ are the frequency detunings while $\Omega_1 = \Omega_a/2$ and $\Omega_2 = \Omega_b/2$ are the effective driving strengths of the two fields. It is now clear that the Hamiltonian in eq. (9) is formally equivalent to that of the TQD described above (*i.e.*, eq. (2)).

Dark state in the TQD. The existence of a dark state in a Λ -type three-level atom in quantum optics is able to suppress absorption or emission when the lower-energy states become degenerate, *i.e.*, $\Delta_1 = \Delta_2$. We demonstrate below that a similar dark state also exists in the TQD and can be used for the ground-state cooling of the NAMR.

After tracing over the degrees of freedom of the electrodes, quantum dynamics of the TQD in the absence of the NAMR is described by

$$\dot{\rho}_d = -i[H_{\text{TQD}}, \rho_d] + \Gamma_1 \mathcal{D}[a_1^\dagger] \rho + \Gamma_2 \mathcal{D}[a_2^\dagger] \rho + \Gamma_3 \mathcal{D}[a_3] \rho_d, \quad (10)$$

where ρ_d is the reduced density matrix of the TQD and Γ_i ($i = 1, 2$ or 3) is the rate for electrons tunneling into or out of the i -th dot. The notation $\mathcal{D}$ for any operator A is given by $\mathcal{D}[A]\rho = A\rho A^\dagger - \frac{1}{2}(A^\dagger A\rho + \rho A^\dagger A)$. Considering equal energy detunings of the two lower-energy states $|1\rangle$ and $|2\rangle$ with respect to the excited state $|3\rangle$, *i.e.*, $\Delta_1 = \Delta_2 = \Delta$, the eigenstates $|g\rangle$, $|-\rangle$ and $|+\rangle$ of the TQD become $|g\rangle = \beta|3\rangle - \frac{\alpha}{\Omega}(\Omega_1|1\rangle + \Omega_2|2\rangle)$, $|-\rangle = \frac{1}{\Omega}(\Omega_2|1\rangle - \Omega_1|2\rangle)$ and $|+\rangle = \alpha|3\rangle + \frac{\beta}{\Omega}(\Omega_1|1\rangle + \Omega_2|2\rangle)$, where $\alpha = \cos(\theta/2)$, $\beta = \sin(\theta/2)$, $\tan\theta = 2\Omega/\Delta$ and $\Omega = \sqrt{\Omega_1^2 + \Omega_2^2}$. The corresponding eigenenergies are $E_g = -\frac{\Delta + \phi}{2}$, $E_- = -\Delta$ and $E_+ = -\frac{\Delta - \phi}{2}$, with $\phi = \sqrt{\Delta^2 + 4\Omega^2}$. For simplicity, we consider equal couplings of the three quantum dots to the corresponding electrodes, *i.e.*, $\Gamma_1 = \Gamma_2 = \Gamma_3 \equiv \Gamma$. Based on the eigenstate basis of the TQD, the equations of motion for the reduced density matrix elements $\rho_{ij} \equiv \langle i|\rho_d|j\rangle$ ($i, j = 0, g, -$ and $+$) of the TQD are obtained from eq. (10) as

$$\begin{aligned} \dot{\rho}_{00} &= -2\Gamma\rho_{00} + \Gamma\beta^2\rho_{gg} + \Gamma\alpha^2\rho_{++} + \Gamma\alpha\beta(\rho_{+g} + \rho_{g+}), \\ \dot{\rho}_{gg} &= \Gamma\alpha^2\rho_{00} - \Gamma\beta^2\rho_{gg} - \frac{\Gamma}{2}\alpha\beta(\rho_{+g} + \rho_{g+}), \\ \dot{\rho}_{--} &= \Gamma\rho_{00}, \\ \dot{\rho}_{++} &= \Gamma\beta^2\rho_{00} - \Gamma\alpha^2\rho_{++} - \frac{\Gamma}{2}\alpha\beta(\rho_{+g} + \rho_{g+}), \\ \dot{\rho}_{+g} &= -i(E_+ - E_g)\rho_{+g} - \frac{\Gamma}{2}\rho_{+g} - \Gamma\alpha\beta\rho_{00} \\ &\quad - \frac{\Gamma}{2}\alpha\beta(\rho_{++} + \rho_{gg}). \end{aligned} \quad (11)$$

Here the diagonal elements ρ_{00} , ρ_{gg} , ρ_{--} and ρ_{++} are the probabilities of TQD in the states $|0\rangle$, $|g\rangle$, $|-\rangle$ and $|+\rangle$, respectively, while the off-diagonal element ρ_{+g} involves the quantum coherence between states $|+\rangle$ and $|g\rangle$. Effective electron tunneling processes through the TQD are described by these equations above. Firstly, an electron can tunnel from the left electrodes into any of the three eigenstates $|g\rangle$, $|-\rangle$ and $|+\rangle$ of an initially empty TQD, with tunneling rates $\Gamma\alpha^2$, Γ and $\Gamma\beta^2$, respectively (see eq. (11)). Note that the total tunneling rate is 2Γ since each of the two tunnel barriers admits electrons with rate Γ . Then, an electron in the eigenstate $|g\rangle$ ($|+\rangle$) can move out of the TQD to the right electrode with a rate $\Gamma\beta^2$ ($\Gamma\alpha^2$) (see eq. (11)). However, no further tunneling occurs if the state $|-\rangle$, which is orthogonal to $|3\rangle$, is occupied. This results from the destructive quantum interference between the transition $|1\rangle \rightarrow |3\rangle$ (*i.e.*, the electron tunneling from state $|1\rangle$ to state $|3\rangle$ in the TQD system) and the transition $|2\rangle \rightarrow |3\rangle$ (*i.e.*, the electron tunneling from $|2\rangle$ to $|3\rangle$). Therefore, an electron can be trapped in the state $|-\rangle$ which is called a dark state [25]. Furthermore, this dark state depends on the initial condition and the probability of having no electron in the TQD (see eq. (11)). Luckily, within the time scale for the ground-state cooling of the NAMR, the dark state will always arise as shown below.

Ground-state cooling of the nanomechanical resonator. –

Quantum dynamics of coupled NAMR-TQD system.

Rather than analyzing directly the energy exchange between the NAMR and the TQD which involves tedious algebra, we apply a canonical transform $U = e^S$ on the whole system, where $S = \eta(a_3^\dagger a_3 + a_1^\dagger a_1)(b - b^\dagger)$. The transformed Hamiltonian is given by

$$\begin{aligned} H &= UH_{\text{total}}U^\dagger \\ &= H_{\text{leads}} + H_{\text{ph}} + \omega_m b^\dagger b - \Delta_1 a_1^\dagger a_1 - \Delta_2 a_2^\dagger a_2 \\ &\quad + (\Omega_1 a_1^\dagger a_3 + \Omega_2 a_2^\dagger a_3 B + \text{H.c.}) \\ &\quad + \sum_q \{\Omega_q b_q^\dagger [b + \eta(a_3^\dagger a_3 + a_1^\dagger a_1)] + \text{H.c.}\} \\ &\quad + \sum_k [\Omega_{1k} a_1^\dagger c_{1k} B^\dagger + \Omega_{2k} a_2^\dagger c_{2k} \\ &\quad + \Omega_{3k} a_3^\dagger c_{3k} B^\dagger + \text{H.c.}], \end{aligned} \quad (12)$$

where we have neglected a small level shift of $\eta^2\omega_m$ to the states $|1\rangle$ and $|3\rangle$ and defined $B = \exp[-\eta(b - b^\dagger)]$. To describe the quantum dynamics of the coupled NAMR-TQD system, we have derived a master equation (under the Born-Markov approximation) by tracing over the degrees of freedom of both the electrodes and the thermal bath. Up to second order in η , the master equation can be written as

$$\begin{aligned} \frac{d\rho}{dt} &= -i\omega_m[b^\dagger b, \rho] - i[H_{\text{TQD}}, \rho] - i[V(b^\dagger - b), \rho] \\ &\quad - i[V'(b^\dagger - b)^2, \rho] + \mathcal{L}_T\rho + \mathcal{L}_D\rho, \end{aligned} \quad (13)$$

where

$$\begin{aligned}
V &= \eta\Omega_2(a_2^\dagger a_3 - a_3^\dagger a_2), \quad V' = \frac{\eta^2}{2}\Omega_2(a_2^\dagger a_3 + a_3^\dagger a_2), \\
\mathcal{L}_T \rho &= \mathcal{L}_T^{(1)} \rho + \mathcal{L}_T^{(2)} \rho, \\
\mathcal{L}_T^{(1)} \rho &= \Gamma \mathcal{D}[a_1^\dagger] \rho + \Gamma \mathcal{D}[a_2^\dagger] \rho + \Gamma \mathcal{D}[a_3] \rho, \\
\mathcal{L}_T^{(2)} \rho &= -\eta^2 \Gamma \mathcal{D}[a_1^\dagger] \rho - \eta^2 \Gamma \mathcal{D}[a_3] \rho + \eta^2 \Gamma (\mathcal{D}[a_1^\dagger b^\dagger] \rho \\
&\quad + \mathcal{D}[a_1^\dagger b] \rho + \mathcal{D}[a_3 b^\dagger] \rho + \mathcal{D}[a_3 b] \rho) \\
&\quad + \eta^2 \Gamma (a_1 a_1^\dagger b^\dagger b \rho + \rho b^\dagger b a_1 a_1^\dagger - a_1^\dagger \rho b^\dagger b a_1 \\
&\quad - a_1^\dagger b^\dagger b \rho a_1 + a_3^\dagger b^\dagger b a_3 \rho + \rho a_3^\dagger b^\dagger b a_3 \\
&\quad - a_3 \rho a_3^\dagger b^\dagger b - b^\dagger b a_3 \rho a_3^\dagger), \\
\mathcal{L}_D \rho &= \gamma[n(\omega_m) + 1] \mathcal{D}[b] \rho + \gamma n(\omega_m) \mathcal{D}[b^\dagger] \rho.
\end{aligned} \tag{14}$$

Here $\mathcal{L}_T$ and $\mathcal{L}_D$ are Liouvillian operators: $\mathcal{L}_T \rho$ accounts for the dissipation due to the electrodes and $\mathcal{L}_D \rho$ the dissipation in the NAMR induced by the thermal bath. Also, γ denotes the decay rate of excitations in the NAMR induced by the thermal bath and $n(\omega_m)$ is the average boson number at frequency ω_m in the thermal bath. Usually, $\gamma \ll \Gamma$ for an NAMR, so $\mathcal{L}_D \rho$ is a higher-order term.

Quantum dynamics of the NAMR. In order to extract the dynamics of NAMR, we trace out the degrees of freedom of the TQD, which can be regarded as part of the environment experienced by the NAMR. This is achieved through adiabatical elimination [21,32] in the limit $\gamma \ll g \ll \omega_m$, corresponding to a weak coupling of the NAMR with the TQD. At zeroth order in η in which NAMR and TQD are decoupled, the dynamics of the whole system is governed by

$$\frac{d\rho}{dt} = \mathcal{L}_0 \rho, \tag{15}$$

where the Liouvillian operator $\mathcal{L}_0$ is defined by $\mathcal{L}_0 \rho \equiv -i\omega_m[b^\dagger b, \rho] - i[H_{\text{TQD}}, \rho] + \mathcal{L}_T^{(1)} \rho$. The steady-state solution to eq. (15) can be expanded [27,32] in the basis of $\rho_{nn'} = |n\rangle\langle n'| \otimes \rho_{ss}$, with $|n\rangle$ being eigenstates of the NAMR Hamiltonian (*i.e.*, $H_R|n\rangle = n\omega_m|n\rangle$) and ρ_{ss} the density operator ρ_d of the TQD in the steady state (which is the solution to eq. (10) when $d\rho_d/dt = 0$). Here $\rho_{nn'}$ are the eigenvectors of $\mathcal{L}_0$: $\mathcal{L}_0|n\rangle\langle n'| \otimes \rho_{ss} = 0$ ($n, n' = 0, 1, \dots$), corresponding to the eigenvalues $\lambda_{nn'} = -i(n - n')\omega_m$. In particular, for the zero eigenvalue $\lambda_0 \equiv \lambda_{nn} = 0$, the eigenvectors are given by $\mathcal{L}_0|n\rangle\langle n| \otimes \rho_{ss} = 0$ ($n = 0, 1, \dots$), which are infinitely degenerate. The zeroth-order Liouville eigenstates with $\lambda_0 = 0$ are connected by the operators $\mathcal{L}_1$ ($\mathcal{L}_1 \rho \equiv -i[V(b^\dagger - b), \rho]$) and $\mathcal{L}_2$ ($\mathcal{L}_2 \rho \equiv -i[V'(b^\dagger - b)^2, \rho] + \mathcal{L}_T^{(2)} \rho + \mathcal{L}_D \rho$) to the subspaces associated with non-zero eigenvalues $\lambda_k \neq 0$ ($k = 1, 2, \dots$). This is due to the coupling between the NAMR and TQD. In the regime $g \ll \omega_m$ (*i.e.*, $\eta \ll 1$), such coupling is weak and can be analyzed using perturbation theory and adiabatic elimination. We define a projection operator $\mathcal{P}$ on the subspace with zero eigenvalue $\lambda_0 = 0$ of $\mathcal{L}_0$ according to

$$\mathcal{P} \rho \equiv \mathcal{P}^{\text{NAMR}} \otimes \mathcal{P}^{\text{TQD}}, \tag{16}$$

where

$$\begin{aligned}
\mathcal{P}^{\text{NAMR}} X &\equiv \sum_n |n\rangle\langle n| \langle n| X |n\rangle, \\
\mathcal{P}^{\text{TQD}} X &\equiv \rho_{ss} \text{Tr}_{\text{TQD}} X.
\end{aligned} \tag{17}$$

Projection of the above master equation (13) gives the master equation for the reduced density matrix μ of the NAMR, up to second order in η ,

$$\begin{aligned}
\dot{\mu} &= -i(\omega_m + \delta_m)[b^\dagger b, \mu] + \frac{1}{2} \{ \gamma[n(\omega_m) + 1] + A_-(\omega_m) \} \\
&\quad \times [2b\mu b^\dagger - (b^\dagger b\mu + \mu b^\dagger b)] + \frac{1}{2} [\gamma n(\omega_m) \\
&\quad + A_+(\omega_m)][2b^\dagger \mu b - (bb^\dagger \mu + \mu bb^\dagger)].
\end{aligned} \tag{18}$$

Here δ_m is the driving-induced shift of the NAMR frequency given by $\delta_m = \text{Im}[G(i\omega_m) + G(-i\omega_m)]$, where $G(s) = -\langle \tilde{V}(s) V(0) \rangle$ with $\tilde{V}(s)$ being the Laplace transform of $V(s)$. In eq. (18), A_+ and A_- are induced by the coupling to the TQD and are given by $A_\pm = 2\text{Re}[G(\pm i\omega_m)] + \eta^2 \Gamma(\rho_{00}^{\text{st}} + \rho_{33}^{\text{st}})$, where ρ_{00}^{st} and ρ_{33}^{st} are steady-state probabilities of the states $|0\rangle$ (empty TQD) and $|3\rangle$ (single electron in dot 3), respectively.

Average phonon number. From master equation (18), the evolution equation for the phonon number probability distribution $p_n = \langle n | \mu | n \rangle$ of the NAMR is obtained as

$$\begin{aligned}
\frac{dp_n}{dt} &= \{ \gamma[n(\omega_m) + 1] + A_- \} [(n+1)p_{n+1} - np_n] \\
&\quad + [\gamma n(\omega_m) + A_+] [np_{n-1} - (n+1)p_n].
\end{aligned} \tag{19}$$

Hence the evolution of the average phonon number, $\langle n \rangle = \sum_n n p_n$, in the NAMR is described by

$$\frac{d\langle n \rangle}{dt} = -(\gamma + W)\langle n \rangle + \gamma n(\omega_m) + A_+, \tag{20}$$

where $W = A_- - A_+$. In order to cool the NAMR, one needs $W > 0$ (*i.e.*, $A_- > A_+$). Consequently, the steady-state average phonon number in the NAMR is

$$n_{\text{st}} = \frac{\gamma n(\omega_m) + A_+}{\gamma + W}. \tag{21}$$

Here the term $\gamma n(\omega_m)$ in the numerator is due to the thermal bath while A_+ results from the scattering processes by the TQD. We assume that the NAMR is initially at equilibrium with the thermal bath, so that its phonon number is initially $n(\omega_m)$. In order to cool down the NAMR significantly, *i.e.*, $n_{\text{st}} \ll n(\omega_m)$, one needs a large cooling rate $W \gg \gamma$. This indeed can be achieved using typical experimental parameters and will be shown below.

The transition rates $A_\pm$ under the condition $\Delta_1 = \Delta_2 \equiv \Delta$ appropriate for the dark state are found to be

$$\begin{aligned}
A_\pm &= \frac{4\eta^2 \Omega_1^2 \Omega_2^2}{\Omega^2} \frac{\omega_m^2 \Gamma}{4[\Omega^2 - \omega_m(\omega_m \pm \Delta)]^2 + \omega_m^2 \Gamma^2} \\
&\quad + \eta^2 \Gamma(\rho_{00}^{\text{st}} + \rho_{33}^{\text{st}}).
\end{aligned} \tag{22}$$

To cool the NAMR, one needs $A_- > A_+$, which is fulfilled either when $\Delta > 0$ and $\Omega < \omega_m$, or when $\Delta < 0$ and $\Omega > \omega_m$. Assuming also $W \gg \gamma$, the steady-state average phonon number in the NAMR is approximately given by $n_{st} \approx \gamma n(\omega_m)/W + n_f$, where

$$n_f \equiv \frac{A_+}{W} = \frac{4[\Omega^2 - \omega_m(\omega_m - \Delta)]^2 + \omega_m^2 \Gamma^2}{16\Delta\omega_m(\omega_m^2 - \Omega^2)}. \quad (23)$$

It is easy to see that n_f reaches a minimum $n_f^{\min} = (\Gamma/4\Delta)^2$, when the term inside the square brackets in the r.h.s. of eq. (23) becomes zero, *i.e.*,

$$\Omega^2 = \omega_m(\omega_m - \Delta), \quad (24)$$

or $\omega_m = (\phi + \Delta)/2$. The optimal cooling condition in eq. (24) can be fulfilled by properly choosing the parameters Ω , ω_m , and Δ . Then, the steady-state average phonon number in the NAMR can be much smaller than unity provided $\Delta \gg \Gamma$, implying that ground-state cooling of the NAMR is possible. The phonon number n_f attainable according to eq. (23) is identical to the previous result for the cooling of trapped atoms via quantum interference [27]. However, solid-state cooling system proposed here has notable advantages such as easy fabrication on a single chip and high controllability. Specifically, all the relevant parameters (*i.e.*, the detuning Δ , the tunneling rate Γ and the interdot coupling strengths Ω_1 and Ω_2) can be controlled by tuning the gate voltages in the TQD. Thus, the optimal cooling condition in eq. (24) can be conveniently fulfilled for a specified frequency ω_m of the NAMR.

The underlying physics of the optimal cooling condition in eq. (24) can be understood more intuitively in the eigenstate basis of the TQD. In the limit $\gamma \ll g \ll \omega_m$ considered here, the TQD arrives quickly at the dark state $|-\rangle$, compared with the dynamic time scale of the NAMR. Therefore, the TQD is practically always maintained in the dark state. The coupling between the NAMR and the TQD excites the electron to the state $|+\rangle$ most readily when the frequency ω_m of the NAMR is equal to the transition frequency $(\phi + \Delta)/2$ between the states $|-\rangle$ and $|+\rangle$, *i.e.*, $\omega_m = (\phi + \Delta)/2$. This corresponds to the transition $|-, n\rangle \rightarrow |+, n-1\rangle$. The excited electron subsequently tunnels to the right electrode, *i.e.*, $|+, n-1\rangle \rightarrow |0, n-1\rangle$. Meanwhile, another electron tunnels into the TQD, and then the dark state returns promptly, *i.e.*, $|0, n-1\rangle \rightarrow |-, n-1\rangle$. This whole process extracts an energy quantum from the NAMR. Therefore, the NAMR will be cooled to the ground state when this cycle repeats, *i.e.*, $|-, n\rangle \rightarrow |+, n-1\rangle \rightarrow |0, n-1\rangle \rightarrow |-, n-1\rangle \rightarrow \dots$, as illustrated in fig. 2. Here we emphasize that the condition of resonance transition for TQD from the state $|-\rangle$ to the state $|+\rangle$ via the NAMR is equivalent to the optimal cooling condition in eq. (24). Although an electron may also relax from the dark state $|-\rangle$ to the ground state $|g\rangle$ by releasing energy to the NAMR, this heating process of

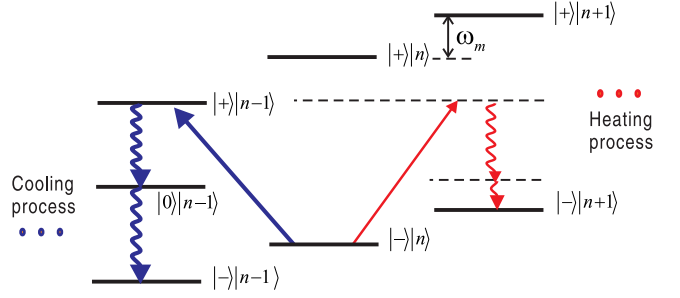


Fig. 2: (Color online) Schematic diagram of the transitions in the coupled NAMR-TQD system, where the cooling process dominates over the heating process. When the frequency of the NAMR is equal to the transition frequency between states $|-\rangle$ and $|+\rangle$, *i.e.*, $\omega_m = (\phi + \Delta)/2$, the process $|-, n\rangle \rightarrow |+, n-1\rangle$ is resonantly enhanced while the process $|-, n\rangle \rightarrow |+, n+1\rangle$ is suppressed. A subsequent tunneling of an electron from the excited state $|+\rangle$ to the right electrode (*i.e.*, $|+, n-1\rangle \rightarrow |0, n-1\rangle$) is followed by another tunneling process with an electron transporting from the left electrode to the dark state $|-\rangle$ (*i.e.*, $|0, n-1\rangle \rightarrow |-, n-1\rangle$), so as to extract a quantum of energy from the NAMR.

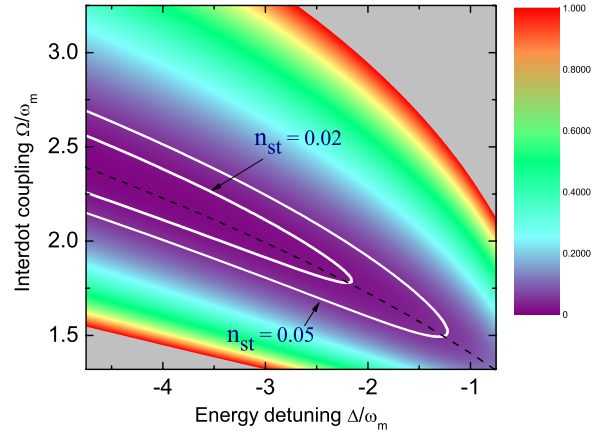


Fig. 3: (Color online) Contour plot of the steady-state average phonon number n_{st} in the NAMR as a function of the normalized driving detuning Δ/ω_m and the interdot coupling Ω/ω_m . The two solid curves correspond to $n_{st} = 0.05$ and 0.02 . The black dashed line represents $\Omega^2 = \omega_m(\omega_m - \Delta)$, under which the NAMR can be optimally cooled. We have chosen $\Omega_1 = \Omega_2 = \Omega/\sqrt{2}$ and typical parameters $\omega_m = 2\pi \times 100$ MHz, $\Gamma = \omega_m$, $Q = 10^5$, $\eta = 0.1$, and $n(\omega_m) = 21$.

the NAMR is strongly suppressed because the frequency of the NAMR is off-resonant to the transition $|-\rangle \rightarrow |g\rangle$ in the TQD.

Figure 3 displays a contour plot of the steady-state average phonon number of the NAMR (n_{st}) as a function of the effective interdot coupling Ω ($= \sqrt{\Omega_1^2 + \Omega_2^2}$) and the energy detuning Δ . Here we choose $\Delta < 0$ and $\Omega > \omega_m$ to make sure that $W > 0$. For these typical parameters, a small $n_{st} < 0.05$ is predicted over a wide range of values on the Ω - Δ plane, which implies that ground-state cooling of the NAMR should be experimentally accessible.

Furthermore, from eq. (22), we obtain a cooling rate $W \approx 2\pi \times 8.52$ MHz by using typical experimental parameters [7,33,34]: $\omega_m = 2\pi \times 100$ MHz, $\Delta = -2\pi \times 40$ GHz and $g = 2\pi \times 10$ MHz, as well as by choosing the interdot couplings $\Omega_1 = \Omega_2 \simeq 2\pi \times 1.41$ GHz to fulfill the optimal cooling condition $\Omega^2 = \omega_m(\omega_m - \Delta)$. Considering an NAMR with a quality factor $Q = 10^5$ (see, *e.g.*, ref. [8]), one has $\gamma = \omega_m/Q = 2\pi \times 1$ kHz. Therefore, appreciable cooling with $W \gg \gamma$ can be achieved. In this case, an NAMR can be cooled from, *e.g.*, an initial temperature $T_m = 100$ mK, corresponding to $n(\omega_m) = 21$, down to $T = 0.8$ mK with $n_{st} = 0.0025$. Starting from an initial temperature T_m , the final temperature of the cooled NAMR should be bound by [35] $T^* = \frac{\omega_m}{|\Delta|} T_m$. In the parameter regime we studied above, $T^* = 0.25$ mK, which is indeed lower than the achieved temperature $T = 0.8$ mK. Moreover, it should be noted that the resolved-sideband cooling condition $\omega_m \gg \Gamma$ is not required in our scheme. Finally, note that in addition to the electrodes coupled to the TQD, there are incoherent processes induced by other degrees of freedom (*e.g.*, background charge fluctuations) in the environment. These processes will affect the dark state of the TQD and then limit the cooling efficiency. Thus, one needs to produce a high-quality TQD with good quantum coherence.

Conclusion. – In summary, we have proposed an approach for achieving the ground-state cooling of an NAMR. It is shown that a dark state, which is decoupled from the excited state, can appear in the TQD in the absence of the NAMR when its two lower-energy localized states become degenerate. With the NAMR capacitively coupled to the TQD and being in resonance with the transition between the dark state and the excited eigenstate in the TQD, we have shown that the ground-state cooling of an NAMR can be achieved in the non-resolved-sideband cooling regime.

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